

A Formal Derivation of the Thick Spray Model from the Enskog–Boltzmann System¹

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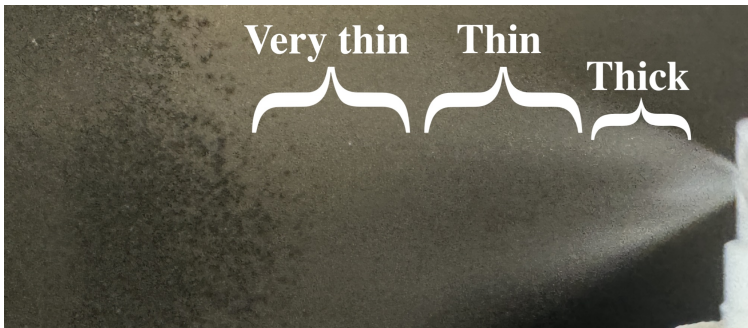
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Motivation



When the droplet volume fraction $\alpha \equiv \alpha(t, x) \in [0, 1]$ is no longer negligible, the gas phase momentum balance becomes

$$\partial_t(\alpha \rho u) + \operatorname{div}_x(\alpha \rho u^{\otimes 2}) + \alpha \nabla_x p = \int_{\mathbb{R}^3} F(v) D(v - u) \, dv,$$

Mathematical Models of Sprays

A spray model is a two-phase flow typically coupling two levels of description:

- **Disperse phase:** droplets, dust, or solid particles, modeled by
 - Hamiltonian or kinetic equations (Vlasov, Boltzmann, Enskog), or
 - fluid-type equations (Stokes, Navier–Stokes, Euler).
- **Gaseous phase:** gas or liquid background, described by
 - kinetic equations (Boltzmann, Enskog), or
 - fluid equations (Stokes, Navier–Stokes, Euler).

This talk will focus on deriving a *kinetic–fluid* description (Vlasov–Euler system) starting from a *kinetic–kinetic* framework (Enskog–Boltzmann system).

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Boltzmann Equation (Dilute Gas)

The Boltzmann equation for hard spheres with radius r_g describes the statistical behavior of a thermodynamic system out of equilibrium by modeling the time evolution of the number density distribution function $f \equiv f(t, x, v) \geq 0$ depending on time $t \in \mathbb{R}_+$, position $x \in \mathbb{R}^3$, and $v \in \mathbb{R}^3$.

$$\partial_t f + v \cdot \nabla_x f = \mathcal{B}[f, f] = \int_{\mathbb{S}^2 \times \mathbb{R}^3} (f(v^o)f(w^o) - f(v)f(w)) \langle v - w, \omega \rangle_+ d\omega dw.$$

Here, the pre-collision velocities (v^o, w^o) are given by

$$\begin{cases} v^o = v^o(v, w, \omega) = v - (v - w) \cdot \omega \omega, \\ w^o = w^o(v, w, \omega) = w + (v - w) \cdot \omega \omega, \end{cases}$$

and $\langle z, \xi \rangle_+ = \max\{\langle z, \xi \rangle, 0\}$ denotes the positive part of the inner product between z and ξ .

Enskog Correction (Finite-Size Particles)

To account for **excluded volume** in binary collisions, we modify the collision integral by shifting by a distance d . The Enskog equation is written as

$$\partial_t f + v \cdot \nabla_x f = \mathcal{E}[f, f],$$

where the (Boltzmann-)Enskog collision integral is written as

$$\mathcal{E}[f, g](x, v) = d^2 \int_{\mathbb{R}^3 \times \mathbb{S}^2} [g(x + d\omega, w^o) f(x, v^o) - g(x - d\omega, w) f(x, v)] \\ (v - w) \cdot \omega H((v - w) \cdot \omega) d\omega dw$$

H denotes the Heaviside step function $H(s) = \begin{cases} 1, & \text{if } s \geq 0, \\ 0, & \text{if } s < 0. \end{cases}$

Dissimilar Binary Hard-Sphere Collision

For collisions between different species, the pre-collision velocities v' and w' are given by

$$\begin{cases} v' = v'(v, w, \omega, \eta) = v - \frac{2\eta}{1+\eta}(v - w) \cdot \omega \omega, \\ w' = w'(v, w, \omega, \eta) = w + \frac{2}{1+\eta}(v - w) \cdot \omega \omega, \end{cases}$$

where η is the mass ratio defined by $\eta = m_g/m_p \leq 1$.

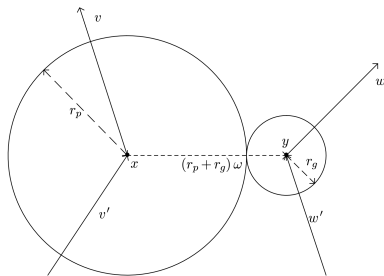


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We consider a binary mixture governed by the coupled kinetic system

$$\begin{cases} (\partial_t + w \cdot \nabla_x) f(t, x, w) = \mathcal{R}[f, F](t, x, w) + \mathcal{C}[f, f](t, x, w), \\ (\partial_t + v \cdot \nabla_x) F(t, x, v) = \mathcal{D}[F, f](t, x, v) + \mathcal{Q}[F, F](t, x, v). \end{cases}$$

Here, $f(t, x, w)$ and $F(t, x, v)$ denote the distribution functions of two species.

Literature on Boltzmann Equations for Gas Mixtures

- P. Degond and B. Lucquin-Desreux, 1996: asymptotics of collision operators with different scales in the space-homogeneous setting
- Y. Guo, 2003: unique global-in-time classical solutions around a global Maxwellian for charged particles
- A. Sotirov and S.-H. Yu, 2010: decay estimates for perturbations when one gas is near vacuum and the other is near a Maxwellian equilibrium state in one space dimension
- C. Bardos and X. Yang, 2012: the boundary-layer problem is well posed
- I. Ampatzoglou, J. K. Miller, and N. Pavlović, 2022: rigorous derivations for mixtures of hard-sphere gases
- Z. Fang and K. Qi, 2025: well-posedness of the scaled equations around the global Maxwellian

Fluid–Kinetic Description for Thin Sprays (Eulerian–Lagrangian Model)

Beyond the kinetic description of collisions, an alternative framework couples a fluid system with a kinetic equation:

$$\left\{ \begin{array}{l} \partial_t \rho + \operatorname{div}_x(\rho u) = 0, \\ \partial_t(\rho u) + \operatorname{div}_x(\rho u^{\otimes 2}) + \nabla_x p = m_g \int_{\mathbb{R}^3} F(v) \Gamma(v - u) dv, \\ \partial_t(\rho E) + \operatorname{div}_x(\rho u E) + \operatorname{div}_x(up) = m_g \int_{\mathbb{R}^3} v \cdot \Gamma(v - u) F(v) dv, \\ \partial_t F + v \cdot \nabla_x F + \operatorname{div}_v[F(v) \Gamma(v - u)] = 0, \\ p = n\theta, \quad E = \frac{|u|^2}{2} + \frac{3\theta}{2m_g}, \quad n = \frac{\rho}{m_g} \end{array} \right.$$

where ρ, u, θ, p, E denote the local density, velocity, temperature, pressure, and total energy of the gas, respectively; $F = F(t, x, v)$ denotes the distribution of particles suspended in the gas; and $\Gamma \equiv \Gamma(v - u)$ is a prescribed drag law describing the friction between the two species.

Fluid–Kinetic Description for Thick Sprays

As the proportion of the dispersed phase increases, the model is modified by introducing the volume fraction α :

$$\left\{ \begin{array}{l} \partial_t(\alpha\rho) + \operatorname{div}_x(\alpha\rho u) = 0, \\ \partial_t(\alpha\rho u) + \operatorname{div}_x(\alpha\rho u^{\otimes 2}) + \alpha\nabla_x p = m_g \int_{\mathbb{R}^3} F(v)\Gamma(v-u)dv, \\ \partial_t(\alpha\rho E) + \operatorname{div}_x(\alpha\rho u E) + \operatorname{div}_x(\alpha p u) + p\partial_t\alpha = m_g \int F(v)\Gamma(v-u) \cdot v dv, \\ \partial_t F + v \cdot \nabla_x F + \operatorname{div}_v \left\{ F(v) \left[\Gamma(v-u) + \frac{4\pi}{3} r_p^3 \nabla_x p \right] \right\} = 0, \\ \rho = m_g n, \quad E = \frac{|u|^2}{2} + \frac{3\theta}{2m_g}, \quad p = n\theta, \quad \alpha = 1 - \frac{4\pi}{3} r_p^3 \int_{\mathbb{R}^3} F(v)dv. \end{array} \right.$$

The thin-spray fluid–kinetic model is recovered in the limiting case $\alpha = 1$.

Thin sprays

- L. Desvillettes, E. Bernard, F. Golse, and V. Ricci, 2017, 2018, 2019: derivation from the Boltzmann equation for binary mixtures
- D. Han-Kwan, A. Moussa, and I. Moyano, 2020: large time behavior of VNS
- R. Danchin, 2026: Fujita–Kato solutions for VNS

Thick Sprays

- B. Després, C. Buet, and L. Desvillettes: linear stability
- B. Després, C. Buet, and V. Fournet, 2024: existence of strong solutions to an averaged thick spray model
- L. Ertzbischoff and D. Han-Kwan, 2025: local-in-time well-posedness

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Main Theorem (Formal Statement)

Let $f^{\eta,\delta}$ and $F^{\eta,\delta}$ be classical solutions of the following Enskog–Boltzmann system:

$$\begin{cases} \partial_t f^{\eta,\delta} + w \cdot \nabla_x f^{\eta,\delta} = \mathcal{E}_1^\eta[f^{\eta,\delta}, F^{\eta,\delta}] + \frac{1}{\delta} \mathcal{B}[f^{\eta,\delta}, f^{\eta,\delta}], \\ \partial_t F^{\eta,\delta} + v \cdot \nabla_x F^{\eta,\delta} = \frac{1}{\eta} \mathcal{E}_2^\eta[F^{\eta,\delta}, f^{\eta,\delta}]. \end{cases}$$

Under the scaling and regularity assumptions, as $\eta, \delta \rightarrow 0$, we have $f^{\eta,\delta} \rightarrow \mathcal{M}[\alpha n, u, \theta/m_g]$ and $F^{\eta,\delta} \rightarrow F$. The limiting quantities F, n, u, θ then solve the following Euler–Vlasov system for thick sprays:

$$\begin{cases} \partial_t(\alpha\rho) + \operatorname{div}_x(\alpha\rho u) = 0, \\ \partial_t(\alpha\rho u) + \operatorname{div}_x(\alpha\rho u^{\otimes 2}) + \alpha\nabla_x p = \int F(v)D(v-u)dv - \operatorname{div}_x\left(\alpha n \int F(v)Q(v-u)dv\right) + o(a^3), \\ \partial_t(\alpha\rho E) + \operatorname{div}_x(\alpha\rho u E) + \operatorname{div}_x(\alpha p u) + p\partial_t\alpha \\ \quad = \int F(v)D(v-u) \cdot v dv - \operatorname{div}_x\left(\alpha n \int F(v)Q(v-u)v dv\right) + o(a^3), \\ \partial_t F + v \cdot \nabla_x F + \operatorname{div}_v\left[F(v)\left(D(v-u) + \frac{4\pi}{3m_g}a^3\nabla_x p\right)\right] = o(a^3), \\ \rho = m_g n, \quad E = \frac{|u|^2}{2} + \frac{3\theta}{2m_g}, \quad p = n\theta, \quad \alpha = 1 - \frac{4\pi}{3}a^3 \int_{\mathbb{R}^3} F(v)dv. \end{cases}$$

Remarks on the result

Q1 Can we eliminate the $o(a^3)$ terms?

A1 No. The leading order of the friction force scales like $O(a^2)$, while buoyancy scales like $O(a^3)$. To retain both effects, the higher-order terms must be kept; otherwise, the friction force would diverge.

Q2 Is $\frac{4\pi}{3}a^3$ a coincidence?

A2 No. The coefficient $\frac{4\pi}{3}a^3$ does not come from directly integrating 1 over a ball. However, in any dimension d , the coefficient in the volume fraction α is exactly the volume of the unit ball in $\mathbb{R}^d$.

Q3 Why are the pressures unbalanced?

A3 This is a feature of non-equilibrium gas-particle systems. It is known as *interfacial pressure*.

Q4 Can we generalize the collision kernel in the Enskog-Boltzmann operator beyond hard-sphere collisions?

A4 Yes, but then the familiar coefficient $\frac{4\pi}{3}a^3$, which is the volume of a sphere of radius a , is no longer exact, since it is specific to the hard-sphere setting.

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Maxwellian in the Gas Limit $\delta \rightarrow 0$

We begin by considering the gas-phase equation

$$\partial_t f^{\eta, \delta} + w \cdot \nabla_x f^{\eta, \delta} = \mathcal{E}_1^\eta[f^{\eta, \delta}, F^{\eta, \delta}] + \frac{1}{\delta} \mathcal{B}[f^{\eta, \delta}, f^{\eta, \delta}]$$

As $\delta \rightarrow 0$, the limit function of gas molecule distribution f should satisfy

$$\mathcal{B}[f, f] = 0.$$

Under suitable assumptions on f , we conclude that f is a Maxwellian distribution, which takes the form of

$$f(t, x, w) = \mathcal{M}[R(t, x), U(t, x), T(t, x)](t, x, w) := \frac{R(t, x)}{(2\pi T(t, x))^{3/2}} \exp\left(-\frac{|w - U(t, x)|^2}{2T(t, x)}\right).$$

Macroscopic quantities

We introduce the following macroscopic quantities for gas phase

$$\rho(t, x) = m_g n(t, x) = \frac{H(\alpha(t, x))}{\alpha(t, x)} \int_{\mathbb{R}^3} m_g f(t, x, w) dw, \quad \text{the local mass density of gas}$$

$$u(t, x) = \frac{H(\alpha(t, x)n(t, x))}{\alpha(t, x)n(t, x)} \int_{\mathbb{R}^3} w f(t, x, w) dw, \quad \text{the bulk velocity of gas}$$

$$\theta(t, x) = \frac{H(\alpha(t, x)n(t, x))}{\alpha(t, x)n(t, x)} \int_{\mathbb{R}^3} m_g \frac{|w - u(t, x)|^2}{3} f(t, x, w) dw, \quad \text{the internal energy of gas}$$

where the scaled volume fraction function $\alpha = \alpha(t, x)$ is defined by

$$\alpha(t, x) := 1 - \frac{4\pi}{3} a^3 \int_{\mathbb{R}^3} F(t, x, v) dv.$$

Then the limit function of $f^{\eta, \delta}(t, x, w)$ should be parametrized by

$$\mathcal{M}[\alpha n, u, \theta/m_g](w).$$

Splitting the collision integral $\mathcal{E}_2^\eta$

Go back to the equation of particles

$$\partial_t F^{\eta,\delta} + v \cdot \nabla_x F^{\eta,\delta} = \frac{1}{\eta} \mathcal{E}_2^\eta[F^{\eta,\delta}, f^{\eta,\delta}].$$

The idea is removing the delocalization from the distribution by

$$f^{\eta,\delta}(x \pm a\omega, w) = f^{\eta,\delta}(x, w) \pm a\omega \cdot \nabla_x f^{\eta,\delta}(x, w) + o(a).$$

Correspondingly, the collision integral $\mathcal{E}_2^\eta$ can be rewritten as

$$\mathcal{E}_2^\eta[F^{\eta,\delta}, f^{\eta,\delta}] = \mathcal{R}_1^\eta[F^{\eta,\delta}, f^{\eta,\delta}] - \mathcal{R}_2^\eta[F^{\eta,\delta}, f^{\eta,\delta}] + o(a^3),$$

Leading friction from $\mathcal{R}_1^\eta$

We consider the weak form of $\mathcal{R}_1^\eta$. For smooth test function $\varphi(v)$, we have

$$\begin{aligned} \int_{\mathbb{R}^3} \mathcal{R}_1^\eta[F^{\eta,\delta}, f^{\eta,\delta}](x, v) \varphi(v) dv \\ &= \int_{\mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{S}^2} [f^{\eta,\delta}(w') F^{\eta,\delta}(v') - f^{\eta,\delta}(w) F^{\eta,\delta}(v)] (v - w) \cdot \omega H((v - w) \cdot \omega) \varphi(v) dv dw d\omega \\ &= \int_{\mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{S}^2} f^{\eta,\delta}(w) F^{\eta,\delta}(v) [\varphi(v') - \varphi(v)] (v - w) \cdot \omega H((v - w) \cdot \omega) dv dw d\omega. \end{aligned}$$

The fundamental theorem of calculus, and show that

$$\varphi(v') - \varphi(v) = (v' - v) \cdot \nabla_v \varphi(v) + o(v' - v) = -\frac{2\eta}{1 + \eta} (v - w) \cdot \omega \omega \cdot \nabla_v \varphi(v) + o(\eta),$$

hence

$$\begin{aligned} \int_{\mathbb{R}^3} \mathcal{R}_1^\eta[F^{\eta,\delta}, f^{\eta,\delta}](x, v) \varphi(v) dv \\ &= -\frac{2a^2\eta}{1 + \eta} \int_{\mathbb{R}^3} \left(F^{\eta,\delta}(x, v) \int_{\mathbb{S}^2} f^{\eta,\delta}(w) ((v - w) \cdot \omega)^2 H((v - w) \cdot \omega) \omega dw d\omega \right) \cdot \nabla_v \varphi(v) dv + o(\eta), \end{aligned}$$

As $\delta \rightarrow 0$, the function $f^{\eta,\delta}$ converges to a Maxwellian distribution. We can compute explicitly the integral

$$\int_{\mathbb{R}^3} (v - w) |v - w| \mathcal{M} \left[\alpha n, u, \frac{\theta}{m_g} \right] (w) dw = \alpha n \frac{\theta}{m_g} q \left(\frac{v - u}{\sqrt{\theta/m_g}} \right),$$

where the function q is given by

$$q(\xi) = \frac{1}{(2\pi)^{3/2}} \int_{\mathbb{R}^3} (\xi - y) |\xi - y| e^{-|y|^2/2} dy.$$

As a consequence, the leading-order friction force reads

$$\lim_{\delta, \eta \rightarrow 0} \int_{\mathbb{R}^3} \mathcal{R}_1^\eta [F^{\eta,\delta}, f^{\eta,\delta}] (x, v) \varphi(v) dv = -\pi a^2 \alpha n \frac{\theta}{m_g} \int_{\mathbb{R}^3} F(v) q \left(\frac{v - u}{\sqrt{\theta/m_g}} \right) \cdot \nabla_v \varphi(v) dv.$$

Pressure Term from Delocalization $\mathcal{R}_2^\eta$

We repeat the same process for $\mathcal{R}_2^\eta$

$$\begin{aligned} & \int_{\mathbb{R}^3} \mathcal{R}_2^\eta[F^{\eta,\delta}, f^{\eta,\delta}](x, v) \varphi(v) dv \\ &= \frac{2a^3\eta}{1+\eta} \int_{\mathbb{R}^3} \left(F^{\eta,\delta}(x, v) \int_{\mathbb{S}^2} \omega \cdot \nabla_x f^{\eta,\delta}(w) ((v-w) \cdot \omega)^2 H((v-w) \cdot \omega) \omega dw \right) \cdot \nabla_v \varphi(v) dv + o(\eta). \end{aligned}$$

Note the following identity:

$$2a^3 \int_{|\omega|=1, \omega \cdot \xi > 0} (\omega \cdot \xi)^2 \omega^{\otimes 2} d\omega = \frac{4\pi}{15} a^3 (2\xi^{\otimes 2} + |\xi|^2 \text{Id}) =: Q(\xi).$$

We can compute it when $f^{\eta,\delta}$ is a Maxwellian distribution by splitting $v - w$ into $v - u$ and $u - w$

$$\begin{aligned} & \int_{\mathbb{R}^3 \times \mathbb{S}^2} \omega \cdot \nabla_x \mathcal{M} \left[\alpha n, u, \frac{\theta}{m_g} \right] (w) ((v-w) \cdot \omega)^2 H((v-w) \cdot \omega) \omega dw d\omega \\ &= \frac{2\pi}{3m_g} \nabla_x (\alpha n \theta) + \frac{2\pi}{15m_g} \left[\text{div}_x \left(2\alpha n (v-u)^{\otimes 2} \right) + \nabla_x \left(\alpha n |v-u|^2 \right) \right]. \end{aligned}$$

We can simplify the pressure term by removing the volume fraction inside the spatial gradient through the approximation

$$\nabla_v \cdot [F \nabla_x (\alpha n \theta)] = \nabla_v \cdot [F \nabla_x (n \theta)] + O(a^3).$$

As a result, the limit of the weak form of $\mathcal{R}_2^\eta$ is

$$\begin{aligned} \lim_{\eta, \delta \rightarrow 0} \frac{1}{\eta} \int_{\mathbb{R}^3} \mathcal{R}_2^\eta[F^{\eta, \delta}, f^{\eta, \delta}](v) \varphi(v) \, dv \\ = \int_{\mathbb{R}^3} F(v) \left(\frac{4\pi}{3m_g} a^3 \nabla_x p + \nabla_x \cdot [\alpha n Q(v - u)] \right) \cdot \nabla_v \varphi(v) \, dv + o(a^3), \end{aligned}$$

where the pressure p is determined by the equation of state $p = n\theta$.

Particle Vlasov-Type Equation

Combining the representation for $\mathcal{R}_1^\eta$ and $\mathcal{R}_2^\eta$, after taking limit $\eta, \delta \rightarrow 0$, we obtain the equation of dispersed phase

$$\partial_t F(t, x, v) + v \cdot \nabla_x F(t, x, v) + \operatorname{div}_v \left[F(t, x, v) \left(D(v - u) + \frac{4\pi}{3m_g} a^3 \nabla_x \rho \right) \right] = o(a^3),$$

where friction force $D(v - u)$ is given by

$$D[\alpha n, u, \theta](v - u) := \pi a^2 \alpha n \frac{\theta}{m_g} q \left(\frac{v - u}{\sqrt{\theta/m_g}} \right) + \operatorname{div}_x [\alpha n Q(v - u)].$$

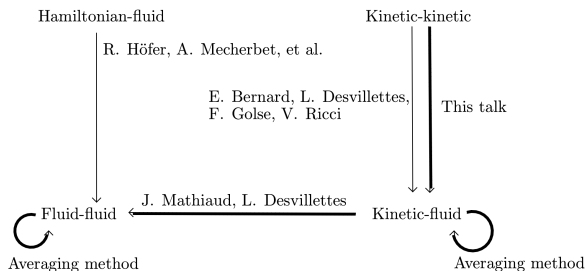
The first term of $D(v - u)$ comes from $\mathcal{R}_1^\eta$ and corresponds to the classical drag term (Desvillettes–Golse–Ricci). The second term comes from the delocalization effects encoded in $\mathcal{R}_2^\eta$.

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Takeaways

- The Euler–Vlasov thick-spray model emerges from the Enskog–Boltzmann system in the regime where the particle-to-gas number ratio is inversely proportional to the mass ratio and gas–gas collisions are sufficiently frequent.
- The derivation provides a kinetic justification for engineering closures and reveals additional $O(a^3)$ effects.



- Polydisperse sprays and more complex settings, including energy exchange and combustion;
- Derivation of fluid–fluid models directly from kinetic–kinetic descriptions, even in multi-species cases;
- Beyond formal limits: rigorous mathematical justifications remain largely open.

Thank you for your attention!